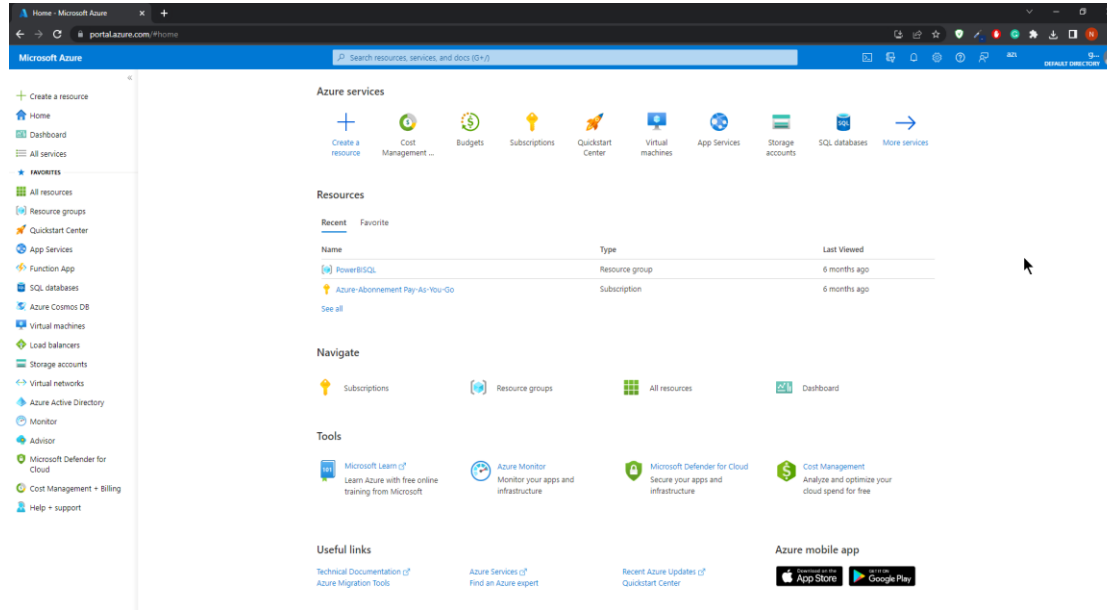


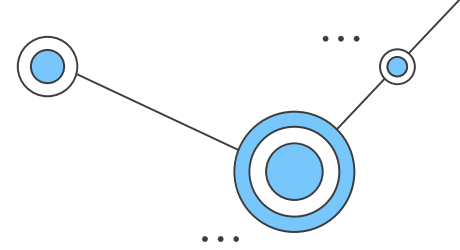
Azure Portal

Browser-based graphical user interface



Can be accessed with any device that has a browser

Azure Portal + Mobile App



Benefits

Graphical interface, easy to learn, easy to manage

Downsides

For bulk tasks it can be tedious

Not ideal for repeatable tasks and automation

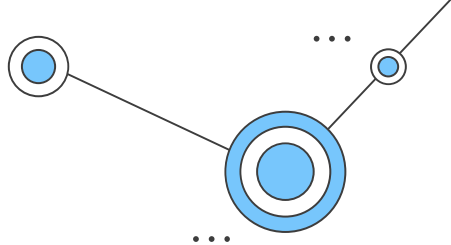
Alternative: Azure CLI and Azure Power Shell

Azure Mobile App

Android + iOS

Limited options but it is mobile

Azure CLI + Azure Power Shell



- *Command-line tools that allow to create and manage resources*
- *Repeatable tasks, bulk creation/management*
- *Cross-plattform, installable on Windows, MacOS and Linux*
- *Overlapping functionality*
- *Main difference: Syntax*
- *Which one to choose: Depending on your previous experience and current work environment*



Azure CLI

Azure CLI is similar to Bash scripting

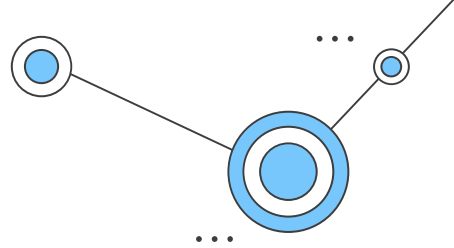
If you mainly work with Linux systems, it feels more familiar

Azure Power Shell

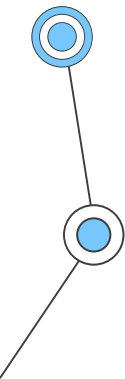
You can use Windows Power Shell for Azure Power Shell

If you mainly work with Windows systems, it's more natural

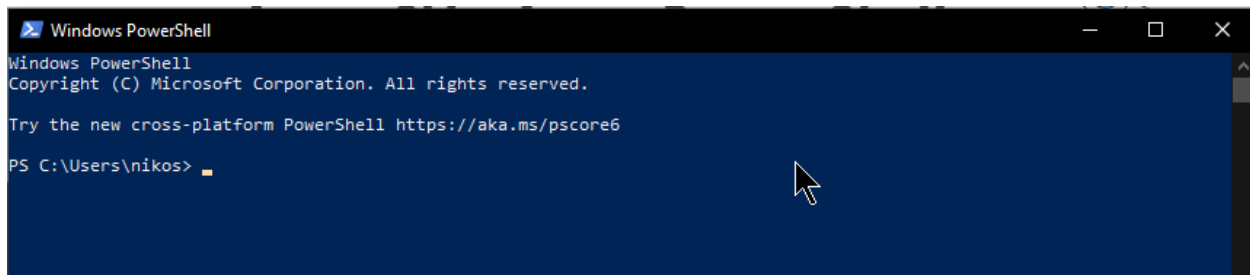
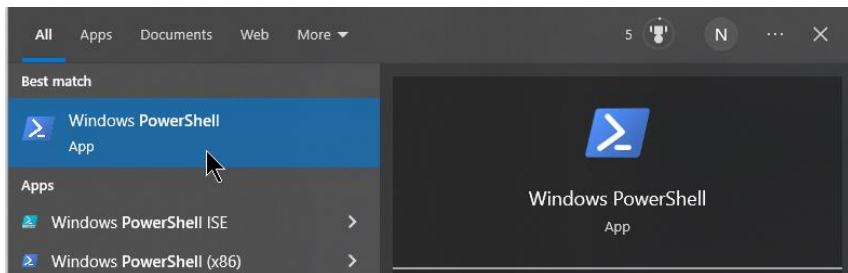
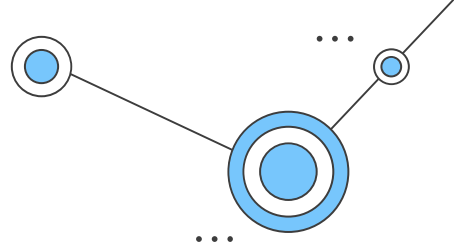
Azure CLI + Azure PowerShell



Command	Azure CLI	Azure PowerShell
Sign in with Web Browser	<code>az login</code>	<code>Connect-AzAccount</code>
List VMs	<code>az vm list</code>	<code>Get-AzVM</code>
Get Help	<code>az --help</code>	<code>Get-Help</code>
List Azure Locations	<code>az account list-locations</code>	<code>Get-AzLocation</code>

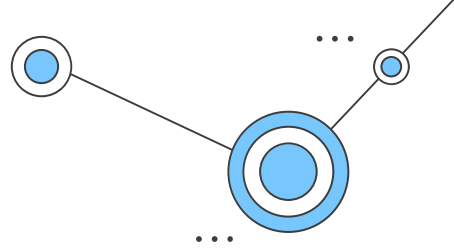


Azure Power Shell Installation



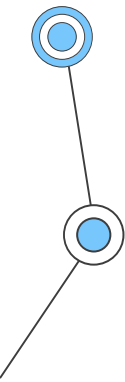
```
Install-Module -Name Az -Scope CurrentUser -Repository PSGallery -Force
```

Azure CLI Installation



Can be installed and run from Windows Command Prompt or PowerShell

Uses `az` commands



Azure Cloud Shell

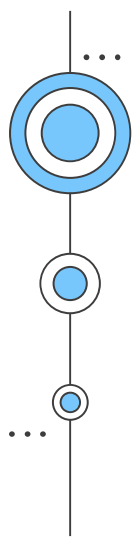
The **easiest way** to use Azure CLI and Azure Power Shell is through the Azure Portal via **Azure Cloud Shell**

You need to mount a dedicated storage account to run

The screenshot displays the Azure Portal interface. The 'Cloud Shell' icon in the top navigation bar is highlighted with a magnifying glass. The main dashboard shows a 'Last Viewed' table with the following data:

Resource	Last Viewed
Resource group	3 days ago
Resource group	3 days ago
Storage account	4 days ago
SQL database	4 days ago
SQL server	4 days ago
Storage account	5 days ago
Resource group	2 weeks ago
Resource group	2 weeks ago
Storage account	4 weeks ago

At the bottom, the 'Welcome to Azure Cloud Shell' dialog is visible, showing the prompt `>_` and the text: 'Select Bash or PowerShell. You can change shells any time via the environment selector in the Cloud Shell toolbar. The most recently used environment will be the default for your next session.' The dialog includes buttons for 'Bash' and 'PowerShell'.

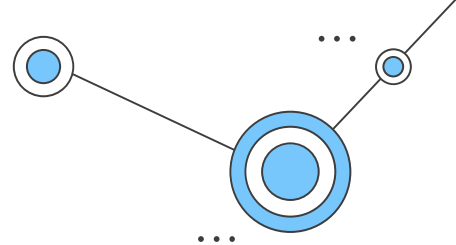


Azure Arc

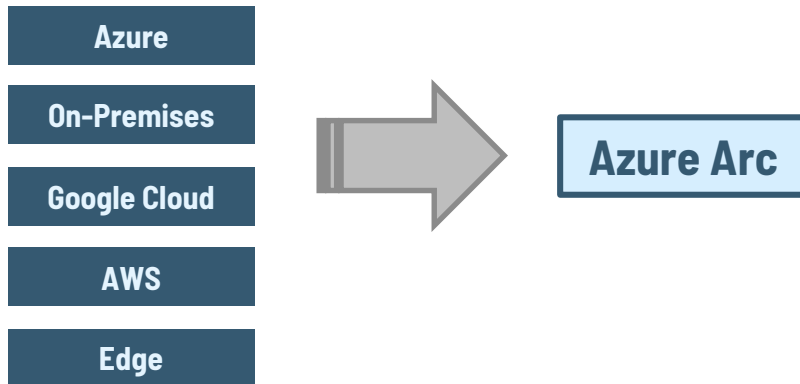


Azure Arc

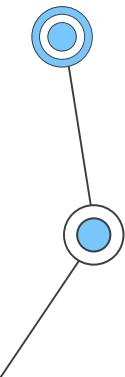
Hybrid and multi-cloud management solution



Managing multiple complex environments can be challenging

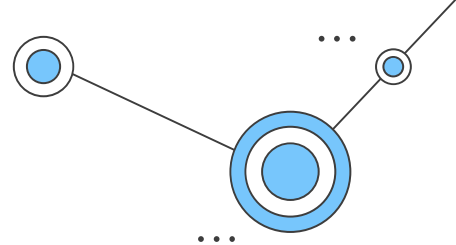


Centralized management platform for multi-cloud, on-premises and edge



Azure Arc

Hybrid and multi-cloud management solution

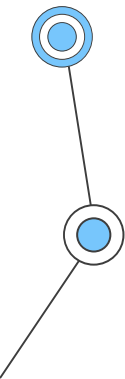


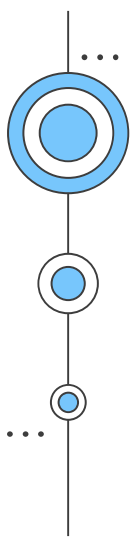
Unified experience:

- Project non-Azure and on-premises resources into Azure (ARM)
- Consistent management, governance and security

Management of the following resources:

- Servers
- Kubernetes clusters
- Azure data services
- SQL Servers
- Virtual machines



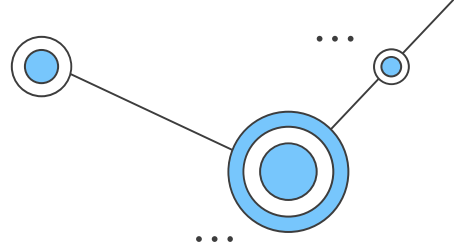


Azure Resource Manager

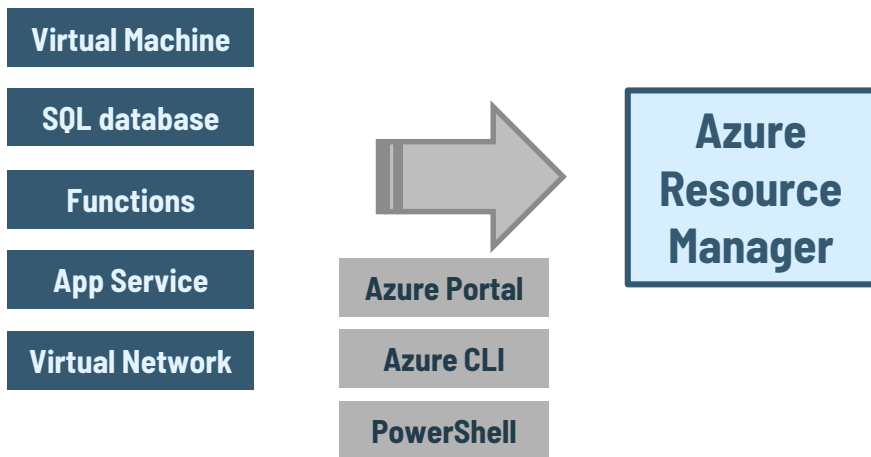


Azure Resource Manager

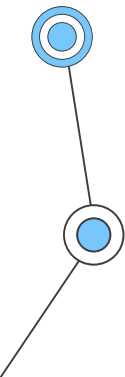
Management layer to create, update, and deploy resources



Whenever you create a resource...



... it will be created through Azure Resource Manager!



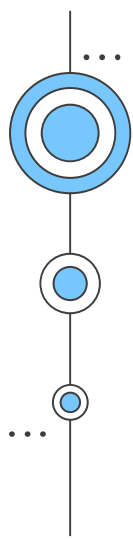
Azure Resource Manager

Management layer to create, update, and deploy resources

ARM templates:

```
1 {  
2   "$schema": "http://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",  
3   "contentVersion": "1.0.0.0",  
4   "parameters": {  
5     "location": {  
6       "type": "string"  
7     },  
8     "networkInterfaceName1": {  
9       "type": "string"  
10    },  
11    "networkSecurityGroupName": {  
12      "type": "string"  
13    },  
14    "networkSecurityGroupRules": [  
15      "type": "array"  
16    ],  
17    "subnetName": {  
18      "type": "string"  
19    },  
20    "virtualNetworkName": {  
21      "type": "string"
```

- Re-deploy existing solutions
- Bulk deployment
- Define dependencies



Summary



Summary

Azure Portal

Web-based graphical interface

Very easy to learn and navigate

Every device with a browsers

Mobile App: Convenient but limited functionality

Azure CLI + PowerShell

Cross-platform command-line tools

Azure CLI scripting similar to Bash (az command)

Bulk deployment and repeatable tasks

Azure Cloud Shell

Accessible through Azure Portal

Access PowerShell and Azure CLI conveniently

Azure Arc

Managing hybrid and multi-cloud

Centralized platform for consistent management, governance and security

Azure Resource Manager

Management layer to create, update, and deploy resources

ARM templates: Re-deployment, bulk-deployment, and define dependencies

